



2024 YEAR 12 ASSESSMENT TASK 2

Mathematics Extension 2

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen.
- Calculators approved by NESA may be used.
- Answer all questions in the booklet provided.
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations.
- A **REFERENCE SHEET** is provided at the back of this paper

Total marks: 69

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II – 59 marks (pages 6 – 9)

- Attempt Questions 11 – 14
- Allow about 1 hour 45 minutes for this section.

Section I

10 marks

Attempt questions 1 - 10

Allow about 15 minutes for this section

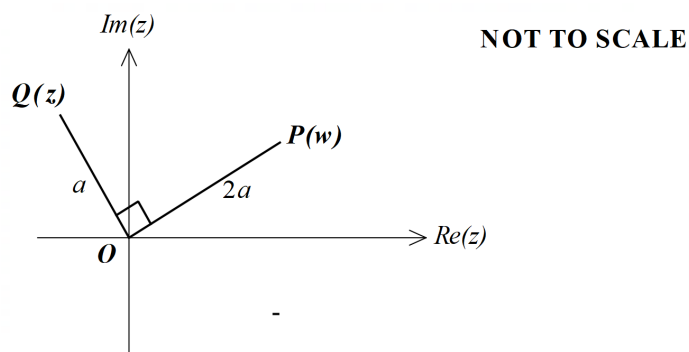
Use the multiple-choice answer page in your booklet to answer Questions 1 – 10.

- 1) A student wants to prove that there is an infinite number of prime numbers. To prove this statement by contradiction, what assumption would the student start their proof with?
- A) There is only one prime number that is even.
- B) There is a finite number of prime numbers.
- C) There are an infinite number of prime numbers.
- D) All prime numbers are less than 100.
- 2) Which of the following is the antiderivative of xe^{2x} ?
- A) $\frac{1}{2}xe^{2x} - \frac{1}{2}e^{2x} + C$
- B) $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$
- C) $\frac{1}{2}x^2e^{2x} - \frac{1}{4}e^{2x} + C$
- D) $\frac{1}{4}x^2e^{2x} + C$
- 3) Which of the following is equivalent to $-2e^{i\theta}$?
- A) $2e^{i(\pi-\theta)}$
- B) $2e^{-i\theta}$
- C) $2e^{i(\theta-\pi)}$
- D) $2e^{i\left(\frac{\pi}{2}-\theta\right)}$

- 4) A quadratic equation with real coefficients has complex roots α and β . What is the possible value of $\text{Arg}(\alpha) + \text{Arg}(\beta)$?

- A) $\frac{\pi}{2}$
- B) 0
- C) π
- D) $\frac{3\pi}{2}$

- 5) In the Argand diagram below the points P and Q represent the complex numbers w and z , respectively where $\angle POQ = 90^\circ$. The distance OP is $2a$ units, and distance OQ is a units. Which of the following is correct?



- A) $w = 2iz$
- B) $w = -2iz$
- C) $w = -\frac{iz}{2}$
- D) $w = -\frac{z}{2i}$

- 6) Which of the following is equivalent to $\int_0^{\frac{\pi}{6}} \sec(2x) \tan(2x) dx$?
- A) $\frac{1}{2}$
- B) $-\frac{1}{2}$
- C) $\frac{1}{4}$
- D) $-\frac{1}{4}$
- 7) The complex number z is defined such that $|z| = |z - 1 - i|$. Which of the following describes the set of values $\text{Arg}(z)$?
- A) $-\pi < \text{Arg}(z) < \pi$
- B) $\text{Arg}(z) = \frac{-3\pi}{4}$ or $\text{Arg}(z) = \frac{\pi}{4}$
- C) $-\frac{\pi}{2} < \text{Arg}(z) < \frac{\pi}{2}$
- D) $-\frac{\pi}{4} < \text{Arg}(z) < \frac{3\pi}{4}$
- 8) What is the negation of the converse of the following statement?
- “If Nathan practises mathematics daily, then he will be successful in the assessment.”
- A) Nathan is successful in the assessment, and he did not practise mathematics daily.
- B) Nathan is unsuccessful in the assessment, and he did not practise mathematics daily.
- C) If Nathan is successful in the assessment, then he did not practise mathematics daily.
- D) If Nathan is unsuccessful in the assessment, then he did not practise mathematics daily.

- 9) Given $a^2 > b^2$, which of the following is always true?
- A) $a + b > 0$
 - B) $a > b$
 - C) $e^{|a|} > e^{|b|}$
 - D) $|a| + |b| > |a + b|$
- 10) What value of k will minimise the integral $I(k) = \int_0^1 (2x^2 - k)^2 dx$?
- A) $k = 0$
 - B) $k = \frac{2}{3}$
 - C) $k = \frac{4}{3}$
 - D) $k = \frac{1}{3}$

End of Section I

Section II (60 marks)**Attempt Questions 11 - 14****Allow about 1 hour 45 minutes for this section**

Answer each question on the appropriate pages of your answer booklet. Extra paper is available.
Your responses should include relevant calculations and/or mathematical reasoning.

Question 11 (15 marks)

- a) Express $2e^{\frac{i\pi}{3}}$ in:
- i) polar form. 1
 - ii) Cartesian form. 1
- b) Prove the following statement is false: 2
- “ $\exists$ a real number x such that $-x^2 + 2x - 5 \geq 0$.”
- c) Evaluate $\int_0^1 \frac{x^2}{1+x^6} dx$. 3
- d) Solve the equation $z^2 = |z|^2 - 4$. 2
- e) If m is a positive integer, prove by contradiction that $\sqrt{4m+6}$ is always irrational. 3
- f) Find $\int \frac{x^2 - 4x + 2}{(x-2)^3} dx$. 3

End of Question 11

Question 12 (14 marks)

a) Prove that for positive integers a, b , if ab is not divisible by 5 then a is not divisible by 5 and b is not divisible by 5. 2

b) If $|z| = 2$ and $w = \frac{z+3}{z}$, find the locus of w . 3

c) Consider three positive real numbers x, y, z .

i) Prove that $x^2 + y^2 \geq 2xy$. 1

ii) Hence, show that $x^2 + y^2 + z^2 \geq xy + yz + zx$. 1

iii) If $x + y + z = 2$, deduce that $xy + yz + zx \leq \frac{4}{3}$. 2

d) Find $\int \frac{dx}{\sqrt{3+4x-x^2}}$. 2

e) Use the substitution $t = \tan \frac{x}{2}$ to find the exact value of $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$. 3

End of Question 12

Question 13 (17 marks)

a) Consider the Statement P:

“ If a number is divisible by 8, then the last three digits of a number is divisible by 8.”

i) Write down the converse of statement P. 1

ii) Show that the converse of statement P is true. 2

b) Consider the complex number $z = \cos \theta + i \sin \theta$

i) Show that $\sin k\theta = \frac{z^k - z^{-k}}{2i}$ where k is an integer. 2

ii) Show that $\cos^2 \theta \sin^2 \theta = -\frac{1}{16} \left(z^2 - \frac{1}{z^2} \right)^2$. 2

iii) Use part ii) to express $\cos^2 \theta \sin^2 \theta$ in the form of $a + b \cos 4\theta$, where a and b are some constants. 2

c) Show that $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{x(\pi - 2x)} = \frac{2}{\pi} \log_e 2$. 4

d) Find $\int \sin^{-1} \sqrt{x} \, dx$. 4

End of Question 13

Question 14 (13 marks)

a) i) Show that $\frac{1}{n^2} < \frac{1}{n-1} - \frac{1}{n}$, where n is positive integer and $n > 1$, **2**

ii) Hence show that for a positive integer x : $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{x^2} < 2$ **2**

b) The terms of a sequence u_n is defined recursively by the formula $u_{n+1} = (1-x)u_n + nx$ for all integers $n \geq 1$, and $u_1 = 0$. Use the principle of mathematical induction to show that for all positive integers n , $u_n = \frac{1}{x} \left(nx - 1 + (1-x)^n \right)$. **3**

c) i) Show that $\frac{(1-t)^n}{t} = t^{n-1} \left(\frac{1}{t} - 1 \right)^n$ **1**

ii) Given that $I_n = \int_1^x \frac{(1-t)^n}{t} dt$, where $n = 1, 2, 3, \dots$, show that $I_n = \frac{(1-x)^n}{n} + I_{n-1}$ **3**

iii) Hence, evaluate I_3 when $x = 2$ **2**

End of Exam

Year-12 Assessment Task 2, 2024 Mathematics Extension-2

Marking Guidelines

Section - I

Multiple Choice (1 - 10)

1 B

6 A

2 B

7 D

3 C

8 A

4 B

9 C

5 B

10 B

Sample Solutions

1) Contradiction : There is finite number of primes. (B)

$$\begin{aligned} 2) \int x e^{2x} dx &= x \frac{e^{2x}}{2} - \int \frac{1}{2} e^{2x} dx \\ &= x \frac{e^{2x}}{2} - \frac{1}{4} e^{2x} + C \end{aligned} \quad (B)$$

$$\begin{aligned} 3) -2e^{i\theta} &= 2(-e^{i\theta}) \\ &= 2(-\cos\theta - i\sin\theta) \\ &= 2(\cos(\theta - \pi) + i\sin(\theta - \pi)) \\ &= 2e^{i(\theta - \pi)} \end{aligned} \quad (C)$$

4) α and β are conjugate of each other

$$\text{If } \alpha = re^{i\theta} \text{ then } \beta = re^{-i\theta}$$

$$\text{Arg } \alpha + \text{Arg } \beta = \theta + (-\theta) = 0 \quad (B)$$

5) $\vec{OQ} = \frac{1}{2} i \vec{OP}$

$$z = \frac{1}{2} i w$$

$$w = \frac{2z}{i} \cdot \frac{i}{i} = -2iz$$

(B)

6) $\int_0^{\pi/6} \sec 2x \tan 2x dx$

Let $2x = u$

$x = 0 \Rightarrow u = 0$

$2dx = du$

$x = \frac{\pi}{6} \Rightarrow u = \frac{\pi}{3}$

$dx = \frac{du}{2}$

$$\begin{aligned} \therefore \frac{1}{2} \int_0^{\pi/3} \sec u \tan u du &= \frac{1}{2} [\sec u]_0^{\pi/3} \\ &= \frac{1}{2} [\sec \frac{\pi}{3} - \sec 0] \\ &= \frac{1}{2} [2 - 1] = \frac{1}{2} \end{aligned}$$

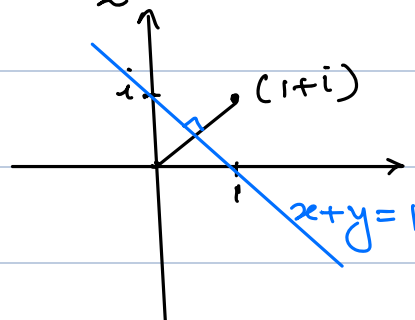
(A)

7) $|z| = |z - 1 - i|$

$$x^2 + y^2 = (x-1)^2 + (y-1)^2$$

$$x^2 + y^2 = x^2 - 2x + 1 + y^2 - 2y + 2$$

$$x + y = 1$$



$$\therefore -\frac{\pi}{4} < \text{Arg}(z) < \frac{3\pi}{4}$$

(D)

8) Let P: Nathan practises mathematics daily.

Q: He will be successful in assessment.

Converse of $P \Rightarrow Q$ is $Q \Rightarrow P$

Negation of $Q \Rightarrow P$ or $\neg(Q \Rightarrow P) \Leftrightarrow Q \wedge \neg P$

$\therefore$ Nathan is successful in assessment, and he did not practise mathematics daily.

(A)

$$9) \quad a^2 > b^2 \Rightarrow a^2 - b^2 > 0$$

$$(a-b)(a+b) > 0$$

$$\therefore a+b > 0 \text{ and } a > b \quad \underline{\text{OR}} \quad a+b < 0 \text{ and } a < b.$$

$\therefore$ Can't be 'A' or 'B'.

Can't be 'D' as $a, b \neq 0$.

$$\text{However } a^2 > b^2 \Rightarrow |a| > |b| \Rightarrow e^{|a|} > e^{|b|} \quad \textcircled{C}$$

$$10) \quad I(k) = \int_0^1 (2x^2 - k)^2 dx$$

Differentiate both sides w.r.t k .

$$\frac{dI}{dk} = \frac{d}{dk} \int_0^1 (2x^2 - k)^2 dx$$

Using the fundamental theorem of Calculus :

$$\begin{aligned} \frac{dI}{dk} &= \int_0^1 \frac{d}{dk} (2x^2 - k)^2 dx \\ &= \int_0^1 -2(2x^2 - k) dx \\ &= -2 \int_0^1 2x^2 - k dx \\ &= -2 \left[\frac{2x^3}{3} - kx \right]_0^1 \\ &= -2 \left(\frac{2}{3} - k \right) \end{aligned}$$

For maximum/minimum value equate $\frac{dI}{dk} = 0$

$$-2 \left(\frac{2}{3} - k \right) = 0 \Rightarrow k = \frac{2}{3}$$

To verify whether $k = \frac{2}{3}$ is minimum or maximum value, test second derivative of $\frac{dI}{dk}$

$$\frac{d^2I}{dk^2} = 2 > 0 \quad \therefore \text{Concave up.}$$

Hence $k = \frac{2}{3}$ is minimum value (B)

(OR)

$$I(k) = \int_0^1 (2x^2 - k)^2 dx \quad \text{--- (1)}$$

$$= \int_0^1 (4x^4 - 4kx^2 + k^2) dx$$

$$= \left[\frac{4x^5}{5} - \frac{4kx^3}{3} + k^2x \right]_0^1$$

$$= \frac{4}{5} - \frac{4k}{3} + k^2$$

Different $I(k)$ w.r.t ' k '

$$\frac{dI}{dk} = -\frac{4}{3} + 2k$$

To minimise ; $\frac{dI}{dk} = 0$

$$-\frac{4}{3} + 2k = 0$$

$$2k = \frac{4}{3}$$

$$k = \frac{2}{3}$$

To verify $k = \frac{2}{3}$ is minimum value, either differentiate $\frac{dI}{dk}$ again w.r.t or substitute $k = \frac{2}{3}$ in $I(k)$ to check whether it is minimum or not.

$\frac{d^2I}{dk^2} = 2 > 0 \quad \therefore$ Concave up, hence $k = \frac{2}{3}$ is minimum value of given integral.

END OF SECTION I

Question 11

a) i) $2e^{i\pi/3} = 2e^{i\pi/3} = 2(\cos \frac{\pi}{3} + i\sin \frac{\pi}{3})$

1 - Correct solution

ii) $2(\frac{1}{2} + i\frac{\sqrt{3}}{2})$ or $1 + i\sqrt{3}$

1 - Correct solution.

b) $\exists x \in \mathbb{R} \text{ s.t. } -x^2 + 2x - 5 \geq 0$

Let $\exists x \in \mathbb{R} \text{ s.t. } -x^2 + 2x - 5 \geq 0$

2 - correct proof

$$-(x^2 - 2x + 1) - 4 \geq 0$$

$$-(x-1)^2 - 4 \geq 0$$

$$-[(x-1)^2 + 4] \geq 0$$

which is false as $(x-1)^2 \geq 0 \forall x \in \mathbb{R}$

1 - applies an appropriate method to prove that the given statement is false.

c) $\int_0^1 \frac{x^2}{1+x^6} dx = \int_0^1 \frac{x^2}{1+(x^3)^2} dx$

Let $x^3 = u$; $x=0 \Rightarrow u=0$ & $x=1 \Rightarrow u=1$

$$3x^2 dx = du$$

$$x^2 dx = \frac{du}{3}$$

$$\therefore \frac{1}{3} \int_0^1 \frac{du}{1+u^2} = \frac{1}{3} [\tan^{-1} u]_0^1 = \frac{1}{3} (\tan^{-1} 1 - \tan^{-1} 0) = \frac{\pi}{12}$$

3 - correct solution

2 - correctly integrate the integral in terms of the substituted variable.

1 - rewrite the integral into an appropriate form by using substitution.

d) Let $z = x + iy$

$$z^2 = |z|^2 - 4$$

$$(x + iy)^2 = x^2 + y^2 - 4$$

$$x^2 - y^2 + 2ixy = x^2 + y^2 - 4$$

$$-y^2 + 2ixy = y^2 - 4$$

Equating real and imaginary parts:

$$-y^2 = y^2 - 4, \quad 2xy = 0$$

$$2y^2 = 4 \quad x = 0 \quad (\because y \neq 0)$$

$$y^2 = 2$$

$$y = \pm\sqrt{2}$$

$$\therefore z = \pm\sqrt{2}i$$

2 - Correct solution.

1 - Create an equation by expressing $z = x + iy$

e) Let $\sqrt{4m+6}$ is rational

$$\therefore \sqrt{4m+6} = \frac{p}{q}, \quad p, q \in \mathbb{Z}^+ \text{ and } p, q \text{ are coprime.}$$

$$4m+6 = \frac{p^2}{q^2}$$

$$2(2m+3) = \frac{p^2}{q^2}$$

$$p^2 = 2(2m+3)q^2 \quad \text{--- (1)}$$

$$\therefore p^2 \text{ is even} \Rightarrow p \text{ is even}$$

Let $p = 2k$, Sub in (1)

$$(2k)^2 = 2(2m+3)q^2$$

$$2k^2 = (2m+3)q^2$$

$$\text{Since } (2m+3) \text{ is odd, } \therefore q^2 = \text{even} \Rightarrow q \text{ is even}$$

which contradict the assumption of p, q have no common factor.

$$\therefore \sqrt{4m+6} \text{ is irrational.}$$

3 - correct proof

2 - Correctly express either 'p' or 'q' as even number by using definition (or) progress towards the proof by showing 'p' and 'q' have common factor.

1 - correctly sets up the proof by contradiction with correct definition.

f)

$$\int \frac{x^2 - 4x + 2}{(x-2)^3} dx$$

$$\int \frac{x^2 - 4x + 4 - 2}{(x-2)^3} dx$$

$$\int \frac{(x-2)^2 - 2}{(x-2)^3} dx$$

$$\int \frac{1}{x-2} - \frac{2}{(x-2)^3} dx$$

$$\int \frac{1}{x-2} dx - 2 \int (x-2)^{-3} dx$$

$$\ln|x-2| - \frac{2(x-2)^{-2}}{(-2)} + C$$

$$\ln|x-2| + \frac{1}{(x-2)^2} + C$$

3 - correct solution
(must have absolute value)

2 - correctly integrate one of the fraction which was decomposed from original fraction.

1 - Correctly express a single fraction into two fractions.

Question-12

a) P: ab is not divisible by 5.

Q: a is not divisible by 5 and b is not divisible by 5.

$$P \Rightarrow Q \iff \neg Q \Rightarrow \neg P.$$

Given statement can be proved by contraposition.

Contraposition statement:

If a or b is divisible by 5 then ab is divisible by 5.

Proof: Let a is divisible by 5

$$a = 5j, \text{ where } j \in \mathbb{Z}^+$$

$$ab = 5jb$$

$$= 5(jb)$$

$$= 5q, \text{ where } q \in \mathbb{Z}^+ \text{ as } j, b \text{ are integers.}$$

$\therefore ab$ is divisible by 5.

Hence the given statement is proved by contraposition.

2 - correct proof.

1 - Correct contraposition
statement

b) $w = \frac{z+3}{z}$

$$wz = z + 3$$

$$wz - z = 3$$

$$z(w-1) = 3$$

$$z = \frac{3}{w-1}$$

Since $|z| = 2$

$$\therefore \left| \frac{3}{w-1} \right| = 2$$

$$\frac{3}{|w-1|} = 2$$

$$|w-1| = \frac{3}{2}$$

$\therefore$ Locus of w is a circle of centre $(1,0)$ and radius $\frac{3}{2}$.

3 - Correct solution

2 - Expressing $|w-1| = \frac{3}{2}$ or

correct approach to find

the locus by using $|z| = 2$

1 - correctly manipulating the

given expression of w and

making z as subject. 'OR'

equivalent approach to find the expression for z .

c) i) $(x-y)^2 \geq 0$

$$x^2 - 2xy + y^2 \geq 0$$

$$x^2 + y^2 \geq 2xy$$

ii) $x^2 + y^2 \geq 2xy$

$$y^2 + z^2 \geq 2yz$$

$$z^2 + x^2 \geq 2zx$$

Add them; $2(x^2 + y^2 + z^2) \geq 2(xy + yz + zx)$

$$x^2 + y^2 + z^2 \geq xy + yz + zx$$

1 - correct proof.

1 - correct proof.

$$\begin{aligned}
 \text{iii)} \quad (x+y+z)^2 &= x^2 + y^2 + z^2 + 2(xy + yz + zx) \\
 &\geq (xy + yz + zx) + 2(xy + yz + zx) \quad (\text{from ii}) \\
 &= 3(xy + yz + zx)
 \end{aligned}$$

$$2^2 \geq 3(xy + yz + zx) \quad (\because x+y+z=2)$$

$$\therefore (xy + yz + zx) \leq \frac{4}{3}$$

2 - correct solution.

1 - correctly using the inequality in part (ii) (or) finding expression for $(x+y+z)^2$

d)

$$\int \frac{dx}{\sqrt{3+4x-x^2}}$$

2 - correct expression.

$$\int \frac{dx}{\sqrt{7-4+4x-x^2}}$$

1 - correctly completing the square in the denominator.

$$\int \frac{dx}{\sqrt{7-(x-2)^2}}$$

$$\sin^{-1} \left(\frac{x-2}{\sqrt{7}} \right) + C$$

e) $\int_0^{\pi/2} \frac{dx}{2+\cos x}$

$$\therefore \int_0^1 \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$t = \tan \frac{x}{2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$x=0 \Rightarrow t=0 ; x=\frac{\pi}{2} \Rightarrow t=1$$

$$\int_0^1 \frac{2dt}{2 + 2t^2 + 1 - t^2}$$

3 - correct solution

$$\int_0^1 \frac{2dt}{3+t^2}$$

2 - Correctly find the integral of $\frac{2dt}{3+t^2}$.

$$2 \int_0^1 \frac{dt}{(\sqrt{3})^2 + t^2}$$

1 - Correctly rewrites the integral in terms of t .

$$\frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$\frac{2}{\sqrt{3}} \left(\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right)$$

$$\frac{2}{\sqrt{3}} \times \frac{\pi}{6} = \frac{\pi}{3\sqrt{3}}$$

Question-13

a) i) Converse of Statement P is:

1 - Correct statement

"If last three digits of a number is divisible by 8 then a number is divisible by 8."

ii) Let the number be N where the last three digits are abc
i.e. $N = \dots abc$

$$\therefore N = 1000 \times Q + abc, \quad Q \in \mathbb{Z}.$$

Since abc is divisible by 8, therefore $abc = 8T, \quad T \in \mathbb{Z}$

$$\therefore N = 1000Q + 8T$$

$$= 8(125Q + T)$$

$$= 8P \quad \text{where } P = 125Q + T \text{ is an integer.}$$

Hence number N is divisible by 8.

Therefore the Converse of statement P is true.

2 - correct proof

1 - correct progress towards the proof by considering the last three digits of a number is multiple of 8 or equivalent progress towards the proof.

b) i) $z = \cos \theta + i \sin \theta$

$$z^k = (\cos \theta + i \sin \theta)^k$$

$$= \cos k\theta + i \sin k\theta \quad \text{--- (1)}$$

$$\frac{1}{z^k} = z^{-k} = (\cos \theta + i \sin \theta)^{-k}$$

$$= \cos(-k\theta) + i \sin(-k\theta)$$

$$= \cos k\theta - i \sin k\theta \quad \text{--- (2)}$$

Add (1) & (2)

$$z^k + \frac{1}{z^k} = 2 \cos k\theta$$

$$\therefore \cos k\theta = \frac{1}{2} \left(z^k + \frac{1}{z^k} \right)$$

Subtract (1) & (2)

$$z^k - \frac{1}{z^k} = 2i \sin k\theta$$

$$\sin k\theta = \frac{1}{2i} \left(z^k - \frac{1}{z^k} \right)$$

ii) $\cos^2 \theta \sin^2 \theta$

$$= (\cos \theta \sin \theta)^2$$

$$= \left[\frac{1}{2} \left(z + \frac{1}{z} \right) \cdot \frac{1}{2i} \left(z - \frac{1}{z} \right) \right]^2$$

$$= \left[\frac{1}{4i} \left(z^2 - \frac{1}{z^2} \right) \right]^2$$

$$= -\frac{1}{16} \left(z^2 - \frac{1}{z^2} \right)^2$$

iii) $\cos^2 \theta \sin^2 \theta = -\frac{1}{16} (2i \sin 2\theta)^2$

$$= \frac{1}{16} (4 \sin^2 2\theta)$$

$$= \frac{1}{8} (1 - \cos 4\theta)$$

$$= \frac{1}{8} - \frac{1}{8} \cos 4\theta$$

$$\therefore a = \frac{1}{8} \text{ and } b = -\frac{1}{8}$$

2 - Correct expression for $\sin k\theta$.

1 - Correct expression for z^k or z^{-k} in term of $\cos \theta$ form or equivalent progress

2 - Correct proof

1 - Correct progress towards the proof.

2 - Correct solution

1 - Using part (ii) to express $z^2 - \frac{1}{z^2} = 2i \sin 2\theta$ or equivalent progress

c) Let $\frac{1}{x(\pi-2x)} = \frac{A}{x} + \frac{B}{\pi-2x}$

$$1 = (\pi-2x)A + Bx$$

Sub $x=0$; $1 = \pi A \Rightarrow A = \frac{1}{\pi}$

Sub $x=1$; $1 = (\pi-2)A + B$

$$1 = 1 - \frac{2}{\pi} + B \quad (\because A = \frac{1}{\pi})$$

$$B = \frac{2}{\pi}$$

$$\begin{aligned} \int_{\pi/6}^{\pi/3} \frac{dx}{x(\pi-2x)} &= \int_{\pi/6}^{\pi/3} \frac{1}{\pi x} + \frac{2}{\pi(\pi-2x)} dx \\ &= \left[\frac{1}{\pi} \ln|x| + \frac{2}{\pi} \frac{\ln|\pi-2x|}{(-2)} \right]_{\pi/6}^{\pi/3} \\ &= \left[\frac{1}{\pi} \ln|x| - \frac{1}{\pi} \ln|\pi-2x| \right]_{\pi/6}^{\pi/3} \\ &= \left[\frac{1}{\pi} \ln \left| \frac{x}{\pi-2x} \right| \right]_{\pi/6}^{\pi/3} \end{aligned}$$

$$= \frac{1}{\pi} \left[\ln \frac{\pi/3}{\pi-2\pi/3} - \ln \frac{\pi/6}{\pi-\pi/3} \right]$$

$$= \frac{1}{\pi} \left[\ln \frac{\pi/3}{\pi/3} - \ln \frac{\pi/6}{2\pi/3} \right]$$

$$= \frac{1}{\pi} \left[\ln 1 - \ln \frac{1}{4} \right]$$

$$= \frac{1}{\pi} (0 - \ln 4^{-1})$$

$$= \frac{1}{\pi} \ln 4$$

$$= \frac{1}{\pi} \ln 2^2 = \frac{2}{\pi} \ln 2$$

4 - Correct proof

3 - Correct integral of both expression and progress towards the substitution.

2 - Correct integral of one expression or equivalent progress.

1 - Decompose the given fraction into two fractions.

ii) $\int \sin^{-1} \sqrt{x} \, dx$

Let $\sqrt{x} = \sin t$

$x = \sin^2 t$

$dx = 2 \sin t \cos t \, dt$

$= \sin 2t \, dt$

$\therefore \int \sin^{-1} \sqrt{x} \, dx = \int t \sin 2t \, dt$

$u = t \quad dv = \sin 2t$

$du = dt \quad v = -\frac{\cos 2t}{2}$

$= -\frac{t \cos 2t}{2} - \int 1 \left(-\frac{\cos 2t}{2}\right) dt$

$= -\frac{1}{2} t \cos 2t + \frac{1}{2} \int \cos 2t \, dt$

$= -\frac{1}{2} t \cos 2t + \frac{1}{4} \sin 2t + C$

$= -\frac{1}{2} t (1 - 2 \sin^2 t) + \frac{1}{4} \cdot 2 \sin t \cos t + C$

$= -\frac{1}{2} \sin^{-1} \sqrt{x} (1 - 2x) + \frac{1}{2} \sin t \sqrt{1 - \sin^2 t} + C$

$= -\frac{1}{2} \sin^{-1} \sqrt{x} + x \sin^{-1} \sqrt{x} + \frac{1}{2} \sqrt{x} \sqrt{1-x} + C$

4 - Correct expression

3 - Correctly integrates and

leave the answer in substitution form (in terms of t or u)

2 - Using integration by parts

1 - appropriate substitution or similar progress.

$$(OR) \int \sin^{-1} \sqrt{x} dx$$

$$u = \sin^{-1} \sqrt{x} \quad v' = 1 \quad (\text{Integration by parts})$$

$$u' = \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}} \quad v = x$$

$$\begin{aligned} \int \sin^{-1} \sqrt{x} dx &= x \sin^{-1} \sqrt{x} - \int \frac{x}{2\sqrt{x}\sqrt{1-x}} dx \\ &= x \sin^{-1} \sqrt{x} - \frac{1}{2} \int \frac{\sqrt{x}}{\sqrt{1-x}} dx \quad \text{--- (1)} \end{aligned}$$

$$\text{Let } I = \int \frac{\sqrt{x}}{\sqrt{1-x}} dx$$

$$\text{Sub } x = \sin^2 \theta \Rightarrow \sqrt{x} = \sin \theta$$

$$dx = 2 \sin \theta \cos \theta d\theta$$

$$I = \int \frac{\sqrt{\sin^2 \theta}}{\sqrt{1-\sin^2 \theta}} (2 \sin \theta \cos \theta) d\theta$$

$$= \int \frac{2 \sin^2 \theta \cos \theta}{\cos \theta} d\theta$$

$$= \int 2 \sin^2 \theta d\theta$$

$$= \int (1 - \cos 2\theta) d\theta$$

$$= \theta - \frac{\sin 2\theta}{2} + C$$

$$= \sin^{-1} \sqrt{x} - \frac{\cancel{2} \sqrt{x} \sqrt{1-x}}{\cancel{2}} + C \quad \left[\because \sin 2\theta = 2 \sin \theta \cos \theta = 2 \sqrt{x} \sqrt{1-x} \right]$$

Sub it back in (1)

$$\int \sin^{-1} \sqrt{x} dx = x \sin^{-1} \sqrt{x} - \frac{1}{2} (\sin^{-1} \sqrt{x} - \sqrt{x} \sqrt{1-x}) + C$$

4 - Correct expression

3 - Correctly integrates and leave the answer in terms of substitution form.

2 - Progress to integrate $\int \frac{\sqrt{x}}{\sqrt{1-x}}$ by using substitution

1 - Correctly using integration by parts

Question - 14

a) i) Since $n > 1$

$$n > n-1$$

$$n^2 > n(n-1)$$

$$\therefore \frac{1}{n^2} < \frac{1}{n(n-1)} \quad (\because n > 1)$$
$$= \frac{1}{n-1} - \frac{1}{n}$$

$$\text{Hence } \frac{1}{n^2} < \frac{1}{n-1} - \frac{1}{n}$$

2 - Correct proof

1 - Correct progress

towards the required proof.

ii) Since $\frac{1}{n^2} < \frac{1}{n-1} - \frac{1}{n}$

$$\text{Sub } n=2; \quad \frac{1}{2^2} < 1 - \frac{1}{2}$$

$$n=3; \quad \frac{1}{3^2} < \frac{1}{2} - \frac{1}{3}$$

$\vdots$

$$n=x \quad \frac{1}{x^2} < \frac{1}{x-1} - \frac{1}{x}$$

2 - Correct proof

1 - Correct approach towards

the proof by using the inequality given in previous part.

Add them:

$$\frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{x^2} < 1 - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} + \dots + \cancel{\frac{1}{x-1}} - \frac{1}{x}$$

Add 1 on both sides

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{x^2} < 1 + 1 - \frac{1}{x}$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{x^2} < 2 - \frac{1}{x} < 2$$

$$\left(\because x > 1 \right. \\ \left. \therefore \frac{1}{x} < 1 \right)$$

Hence proved.

b) Prove true for $n=1$.

$$L.H.S = u_1 = 0$$

$$\begin{aligned} R.H.S &= \frac{1}{x} [x - 1 + (1-x)^1] \\ &= \frac{1}{x} [\cancel{x} - 1 + 1 - \cancel{x}] = 0 \end{aligned}$$

$\therefore$ True for $n=1$.

Assume true for $n=k$.

$$\text{i.e. } u_k = \frac{1}{x} [kx - 1 + (1-x)^k], \quad k \geq 1$$

Prove true for $n=k+1$.

$$\text{i.e. } u_{k+1} = \frac{1}{x} [(k+1)x - 1 + (1-x)^{k+1}]$$

Proof: $L.H.S = u_{k+1}$

$$= (1-x)u_k + kx \quad (\text{Using recursive formula})$$

$$= (1-x) \cdot \frac{1}{x} [kx - 1 + (1-x)^k] + kx \quad (\text{Using assumption})$$

$$= \frac{1}{x} [(1-x)kx - (1-x) + (1-x)^{k+1} + kx^2]$$

$$= \frac{1}{x} [kx - \cancel{kx^2} - 1 + x + (1-x)^{k+1} + \cancel{kx^2}]$$

$$= \frac{1}{x} [(k+1)x - 1 + (1-x)^{k+1}]$$

$$= R.H.S$$

$\therefore$ True for $n=k+1$ if it is assumed to be true for $n=k$.

Since $n=1$ is shown true, by the principle of induction, it is true for all positive integers n .

3 - Correct Solution

2 - attempts to use the assumption to prove the required result.

1 - Correctly proves the base case.

$$\begin{aligned}
 \text{c) i)} \quad \frac{(1-t)^n}{t} &= \frac{1}{t} \left[t \left(\frac{1}{t} - 1 \right) \right]^n \\
 &= \frac{1}{t} \cdot t^n \left(\frac{1}{t} - 1 \right)^n \\
 &= t^{n-1} \left(\frac{1}{t} - 1 \right)^n
 \end{aligned}$$

1 - correct proof

$$\begin{aligned}
 \text{ii)} \quad I_n &= \int_1^x \frac{(1-t)^n}{t} dt \\
 &= \int_1^x t^{n-1} \left(\frac{1}{t} - 1 \right)^n dt
 \end{aligned}$$

3 - correct solution

2 - correct substitution and establishes reduction formula

Integrate by parts:

1 - correct integration by parts

$$u = \left(\frac{1}{t} - 1 \right)^n$$

$$dv = t^{n-1}$$

$$du = n \left(\frac{1}{t} - 1 \right)^{n-1} \left(-\frac{1}{t^2} \right)$$

$$v = \frac{t^n}{n}$$

$$\therefore I_n = \left[\frac{t^n}{n} \left(\frac{1}{t} - 1 \right)^n \right]_1^x - \int_1^x n \left(\frac{1}{t} - 1 \right)^{n-1} \left(-\frac{1}{t^2} \right) \frac{t^n}{n} dt$$

$$= \left[\frac{x^n}{n} \left(\frac{1}{x} - 1 \right)^n - \frac{1}{n} (1-1)^n \right] + \int_1^x t^{n-2} \left(\frac{1}{t} - 1 \right)^{n-1} dt$$

$$= \frac{x^n}{n} \left(\frac{1}{x} - 1 \right)^n + I_{n-1}$$

$$= \frac{x^n (1-x)^n}{x^n \cdot n} + I_{n-1}$$

$$= \frac{(1-x)^n}{n} + I_{n-1}$$

iii) When $x = 2$;

$$I_3 = \frac{(1-2)^3}{3} + I_2 = -\frac{1}{3} + I_2$$

$$I_2 = \frac{(1-2)^2}{2} + I_1 = \frac{1}{2} + I_1$$

$$I_1 = \frac{(1-2)^1}{1} + I_0 = -1 + I_0$$

$$\begin{aligned}\therefore I_3 &= -\frac{1}{3} + \frac{1}{2} - 1 + I_0 \\ &= -\frac{5}{6} + I_0\end{aligned}$$

$$I_0 = \int_1^2 \frac{1}{t} dt = [\ln t]_1^2 = \ln 2 - \ln 1 = \ln 2$$

$$\therefore I_3 = -\frac{5}{6} + \ln 2.$$

2 - Correct value of I_3 .

1 - Correctly applies the reduction formula once.
(Or) Correctly find I_0 .

END OF EXAM